ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** If a number is divisible by 3, then it is divisible by 6.	Interchange the hypothesis and the conclusion.
2	**(d)** False	Counterexample: 9 is divisible by 3 but not by 6.
3	**(d)** $a = -3,\ b = 2$	$a^2 = 9 > b^2 = 4$.
4	**(d)** 9	List the factors of 36 in pairs; the count is 9 (odd, so 36 is a perfect square).
5	**(c)** right-angled	$9^2 + 40^2 = 1681$ and $41^2 = 1681$.
6	**(a)** 10	$n = 10$ gives 121, which is composite.
7	**(b)** a counterexample	One counterexample is enough.
8	**(a)** 25	Only a perfect square has an odd number of factors; $25 = 5^2$.
9	**(a)** If a quadrilateral is a rectangle, then it is a square.	Interchange the hypothesis and the conclusion.
10	**(b)** False	Counterexample: a 4 cm by 6 cm rectangle is not a square.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Converse: If a number is divisible by 3, then it is divisible by 6. (1); it is false, counterexample: 9 is divisible by 3 but not by 6 (1).	1+1
12	Take 2 and 3 (1); $2 + 3 = 5$ is odd (1).	1+1
13	64 has 7 factors (1); odd number, so it is a perfect square (1).	1+1
14	Converse: If a quadrilateral is a rectangle, then it is a square. (1); it is false, counterexample: a 4 cm by 6 cm rectangle is not a square (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Converse: If a number is divisible by 3, then it is divisible by 6. (1); Counterexample: 9 is divisible by 3 but not by 6 (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
16	A false statement needs only one counterexample (1); Take 2 (1); 2 is prime and even, so the statement is false (1).	1+1+1
17	Longest side 39; $15^2 + 36^2 = 1521$ (1); $39^2 = 1521$ (1); Right-angled (1).	1+1+1
	OR: Converse: If a quadrilateral is a rectangle, then it is a square. (1); Counterexample: a 4 cm by 6 cm rectangle is not a square (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
18	A false statement needs only one counterexample (1); Take $a = -3,\ b = 2$ (1); $a^2 = 9 > b^2 = 4$, so the statement is false (1).	1+1+1
	OR: Longest side 34; $16^2 + 30^2 = 1156$ (1); $34^2 = 1156$ (1); Right-angled (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let the integers be $n, n+1, \ldots, n+2$ (1); Sum $= 3n + 3 = 3(n + 1)$ (1); A multiple of 3, proved (1); Check: $8 + 9 + 10 = 27 = 3 \times 9$ (2).	1+1+1+2
	OR: Let the integer be $2m + 1$; its square $= 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1$ is odd (1); Converse: if n^2 is odd then n is odd (1); For $7^2 = 49$ (odd), n is odd (1); If n were even, n^2 would be even, so the converse is true (2).	1+1+1+2
20	Let the integers be $n, n+1, \ldots, n+4$ (1); Sum $= 5n + 10 = 5(n + 2)$ (1); A multiple of 5, proved (1); Check: $12 + 13 + 14 + 15 + 16 = 70 = 5 \times 14$ (2).	1+1+1+2
	OR: Let the integer be $2m + 1$; its square $= 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1$ is odd (1); Converse: if n^2 is odd then n is odd (1); For $11^2 = 121$ (odd), n is odd (1); If n were even, n^2 would be even, so the converse is true (2).	1+1+1+2

Mistakes to avoid:

- Swapping the converse incorrectly, e.g. writing 'If not P, then not Q', which is a different statement.
- Assuming the converse of a true statement is also true; always test it with an example.
- Trying to 'prove' a general statement by checking a few examples; examples can only disprove it.
- Giving a counterexample that does not satisfy the hypothesis; it must make the hypothesis true and the conclusion false.
- Confusing hypothesis and conclusion when the statement is written in a form like 'Q, if P'.
- Writing a proof without defining the objects, e.g. not writing odd numbers as $2k + 1$ before multiplying.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $n = 40$	$1600 + 40 + 41 = 1681 = 41^2$.
2	**(d)** 10	List the factors of 48 in pairs; the count is 10 (even, so 48 is not a perfect square).
3	**(d)** right-angled	$10^2 + 24^2 = 676$ and $26^2 = 676$.
4	**(c)** 16	$n = 16$ gives 289, which is composite.
5	**(c)** do not prove it for all cases	Examples are not proofs.
6	**(c)** 64	Only a perfect square has an odd number of factors; $64 = 8^2$.
7	**(a)** If the diagonals of a quadrilateral are perpendicular, then it is a rhombus.	Interchange the hypothesis and the conclusion.
8	**(d)** False	Counterexample: a kite that is not a rhombus has perpendicular diagonals.
9	**(d)** $x = -4$	$16 > 9$ but $-4 < 3$.
10	**(d)** 3	List the factors of 49 in pairs; the count is 3 (odd, so 49 is a perfect square).

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Take $n = 40$ (1); $1600 + 40 + 41 = 1681 = 41^2$ (1).	1+1
12	72 has 12 factors (1); even number, so it is not a perfect square (1).	1+1
13	Converse: If a triangle is isosceles, then it is equilateral. (1); it is false, counterexample: a triangle with sides $5, 5, 8$ is isosceles but not equilateral (1).	1+1
14	Take $\sqrt{2} \times \sqrt{2}$ (1); $\sqrt{2} \times \sqrt{2} = 2$ is rational (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Converse: If a triangle is isosceles, then it is equilateral. (1); Counterexample: a triangle with sides $5, 5, 8$ is isosceles but not equilateral (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
16	A false statement needs only one counterexample (1); Take 9 (1); $9 = 3 \times 3$, so the statement is false (1).	1+1+1
17	Longest side 85; $13^2 + 84^2 = 7225$ (1); $85^2 = 7225$ (1); Right-angled (1).	1+1+1
	OR: Converse: If $x^2 = 9$, then $x = 3$. (1); Counterexample: $x = -3$ gives $x^2 = 9$ but $x \neq 3$ (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
18	A false statement needs only one counterexample (1); Take $n = 7$ (1); $2(7) + 1 = 15 = 3 \times 5$, so the statement is false (1).	1+1+1
	OR: Longest side 50; $14^2 + 48^2 = 2500$ (1); $50^2 = 2500$ (1); Right-angled (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let the integers be $n, n+1, \ldots, n+6$ (1); Sum $= 7n + 21 = 7(n+3)$ (1); A multiple of 7, proved (1); Check: $4 + 5 + 6 + 7 + 8 + 9 + 10 = 49 = 7 \times 7$ (2).	1+1+1+2
	OR: Let the integer be $2m + 1$; its square $= 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1$ is odd (1); Converse: if n^2 is odd then n is odd (1); For $15^2 = 225$ (odd), n is odd (1); If n were even, n^2 would be even, so the converse is true (2).	1+1+1+2
20	Let the integers be $n, n+1, \ldots, n+2$ (1); Sum $= 3n + 3 = 3(n+1)$ (1); A multiple of 3, proved (1); Check: $25 + 26 + 27 = 78 = 3 \times 26$ (2).	1+1+1+2
	OR: Let the integer be $2m + 1$; its square $= 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1$ is odd (1); Converse: if n^2 is odd then n is odd (1); For $9^2 = 81$ (odd), n is odd (1); If n were even, n^2 would be even, so the converse is true (2).	1+1+1+2

Mistakes to avoid:

- Swapping the converse incorrectly, e.g. writing 'If not P, then not Q', which is a different statement.

- Assuming the converse of a true statement is also true; always test it with an example.
- Trying to 'prove' a general statement by checking a few examples; examples can only disprove it.
- Giving a counterexample that does not satisfy the hypothesis; it must make the hypothesis true and the conclusion false.
- Confusing hypothesis and conclusion when the statement is written in a form like 'Q, if P'.
- Writing a proof without defining the objects, e.g. not writing odd numbers as $2k + 1$ before multiplying.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Propositions and their Converses** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** not right-angled	$11^2 + 60^2 = 3721$ and $62^2 = 3844$.
2	**(d)** 4	$n = 4$ gives 25, which is composite.
3	**(a)** P if and only if Q	Both directions hold.
4	**(c)** 36	Only a perfect square has an odd number of factors; $36 = 6^2$.
5	**(c)** If a number is even, then it is a multiple of 4.	Interchange the hypothesis and the conclusion.
6	**(a)** False	Counterexample: 6 is even but not a multiple of 4.
7	**(b)** $\sqrt{2} \times \sqrt{2}$	$\sqrt{2} \times \sqrt{2} = 2$ is rational.
8	**(b)** 9	List the factors of 100 in pairs; the count is 9 (odd, so 100 is a perfect square).
9	**(b)** right-angled	$12^2 + 35^2 = 1369$ and $37^2 = 1369$.
10	**(b)** 40	$n = 40$ gives 1681, which is composite.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	81 has 5 factors (1); odd number, so it is a perfect square (1).	1+1
12	Converse: If a number is odd, then it is prime and greater than 2. (1); it is false, counterexample: 9 is odd but not prime (1).	1+1
13	Take $n = 11$ (1); $121 - 11 + 11 = 121 = 11^2$ (1).	1+1
14	90 has 12 factors (1); even number, so it is not a perfect square (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Converse: If a number is odd, then it is prime and greater than 2. (1); Counterexample: 9 is odd but not prime (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
16	A false statement needs only one counterexample (1); Take an isosceles trapezium (1); An isosceles trapezium has equal diagonals but is not a rectangle, so the statement is false (1).	1+1+1
17	Longest side 31; $18^2 + 24^2 = 900$ (1); $31^2 = 961$ (1); Not right-angled (1).	1+1+1
	OR: Converse: If a number is a multiple of 4, then it is a multiple of 8. (1); Counterexample: 12 is a multiple of 4 but not of 8 (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
18	A false statement needs only one counterexample (1); Take $x = -1,\ y = -4$ (1); $1 < 16$, so the statement is false (1).	1+1+1
	OR: Longest side 54; $28^2 + 45^2 = 2809$ (1); $54^2 = 2916$ (1); Not right-angled (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let the integers be $n, n+1, \ldots, n+4$ (1); Sum $= 5n + 10 = 5(n+2)$ (1); A multiple of 5, proved (1); Check: $31 + 32 + 33 + 34 + 35 = 165 = 5 \times 33$ (2).	1+1+1+2
	OR: Let the integer be $2m+1$; its square $= 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1$ is odd (1); Converse: if n^2 is odd then n is odd (1); For $13^2 = 169$ (odd), n is odd (1); If n were even, n^2 would be even, so the converse is true (2).	1+1+1+2
20	Let the integers be $n, n+1, \ldots, n+8$ (1); Sum $= 9n + 36 = 9(n+4)$ (1); A multiple of 9, proved (1); Check: $6 + 7 + 8 + 9 + 10 + 11 + 12 + 13 + 14 = 90 = 9 \times 10$ (2).	1+1+1+2
	OR: Let the integer be $2m+1$; its square $= 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1$ is odd (1); Converse: if n^2 is odd then n is odd (1); For $21^2 = 441$ (odd), n is odd (1); If n were even, n^2 would be even, so the converse is true (2).	1+1+1+2

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